

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from sklearn.datasets import load_iris
import warnings
warnings.filterwarnings("ignore")
```

```
In [2]: data = load_iris()
```

```
In [3]: print("Features and their types:")
print(data.keys())
```

Features and their types:
dict_keys(['data', 'target', 'frame', 'target_names', 'DESCR', 'feature_names', 'filename', 'data_module'])

```
In [4]: data.feature_names
```

```
Out[4]: ['sepal length (cm)',
'sepal width (cm)',
'petal length (cm)',
'petal width (cm)']
```

```
In [5]: df = pd.DataFrame()
df[data['feature_names']] = data['data']
df['label'] = data['target']
df
```

```
Out[5]:
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	label
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	2
146	6.3	2.5	5.0	1.9	2
147	6.5	3.0	5.2	2.0	2
148	6.2	3.4	5.4	2.3	2
149	5.9	3.0	5.1	1.8	2

150 rows × 5 columns

```
In [6]: df.head()
```

```
Out[6]:
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	label
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

```
In [7]: df.info()
```

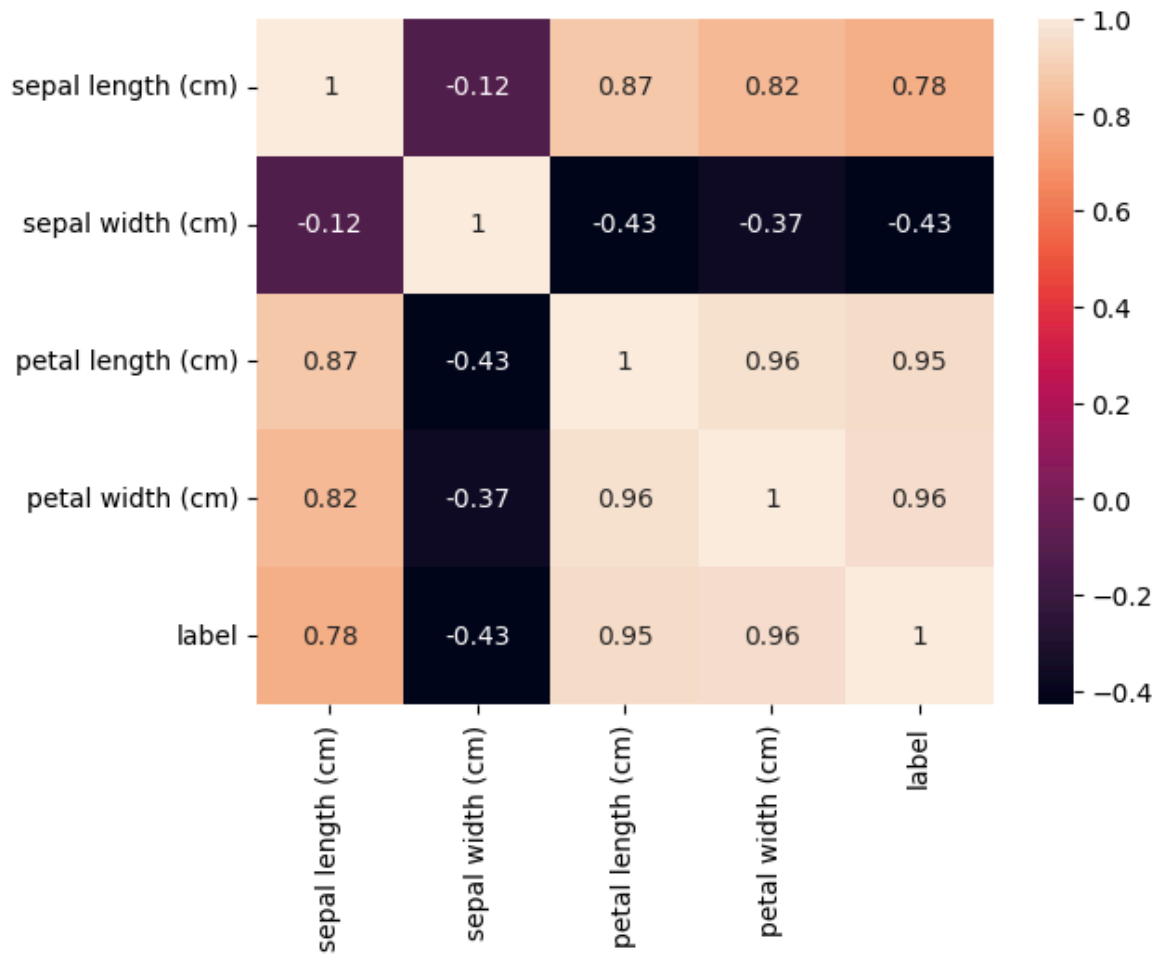
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   sepal length (cm)      150 non-null    float64
1   sepal width (cm)       150 non-null    float64
2   petal length (cm)      150 non-null    float64
3   petal width (cm)       150 non-null    float64
4   label                  150 non-null    int64
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
```

```
In [8]: df.describe()
```

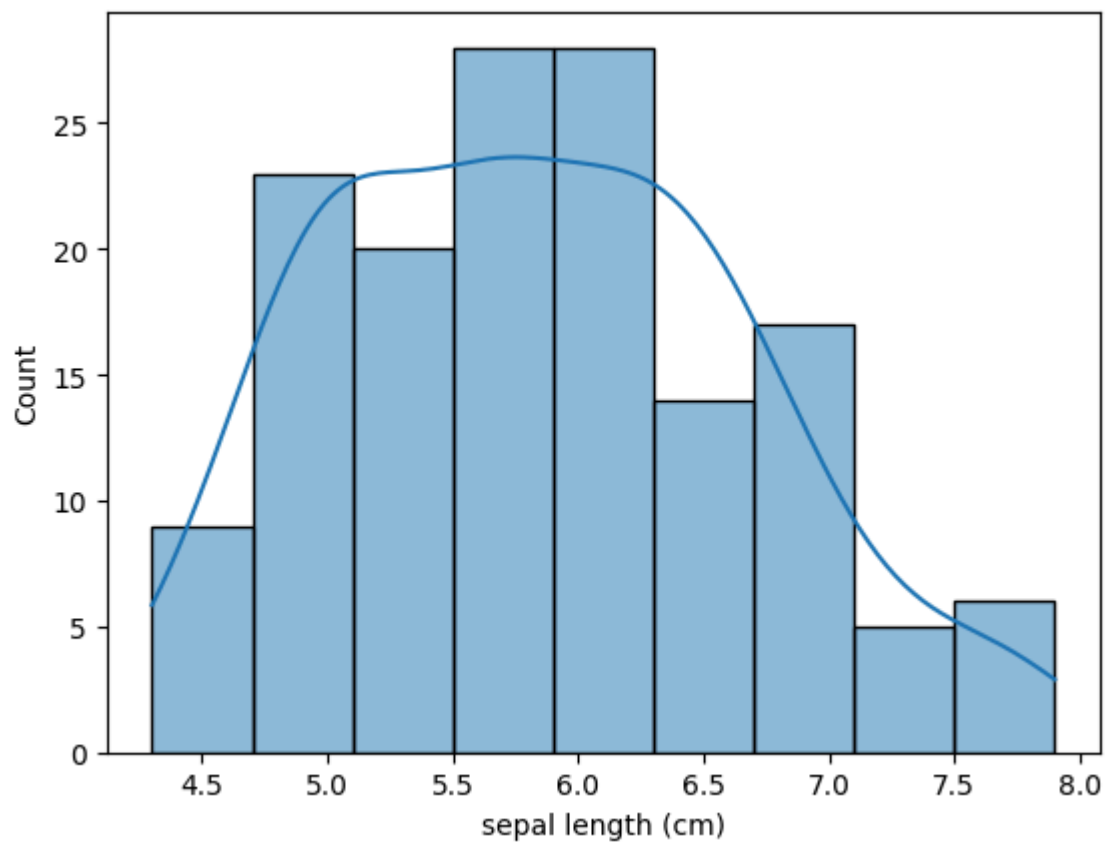
```
Out[8]:
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	label
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333	1.000000
std	0.828066	0.435866	1.765298	0.762238	0.819232
min	4.300000	2.000000	1.000000	0.100000	0.000000
25%	5.100000	2.800000	1.600000	0.300000	0.000000
50%	5.800000	3.000000	4.350000	1.300000	1.000000
75%	6.400000	3.300000	5.100000	1.800000	2.000000
max	7.900000	4.400000	6.900000	2.500000	2.000000

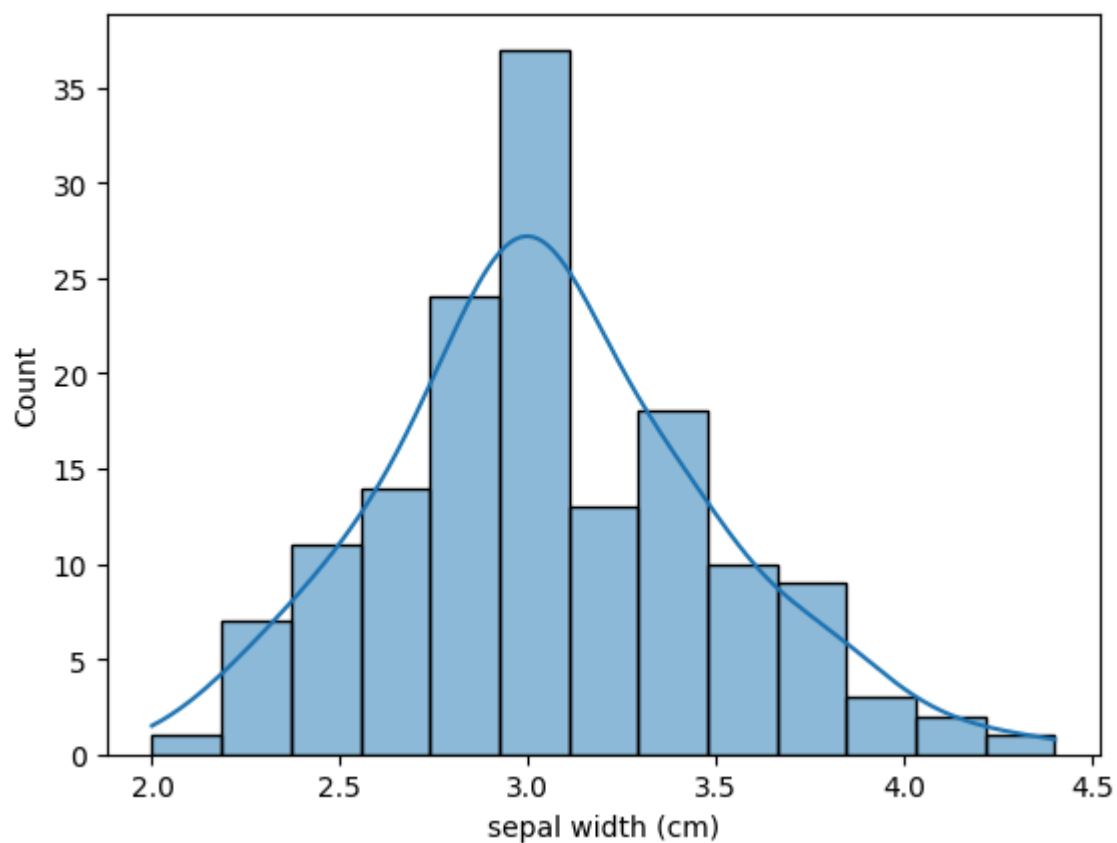
```
In [9]: sns.heatmap(df.corr(), annot=True)
plt.show()
```



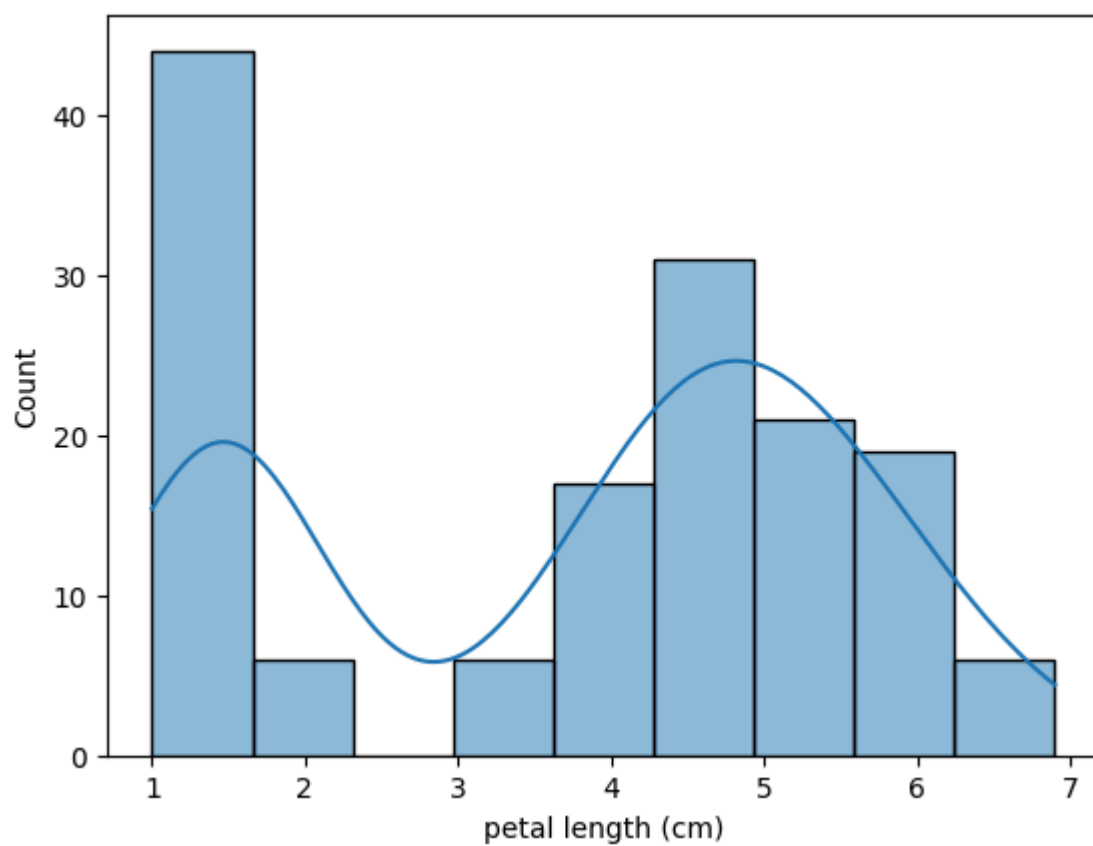
```
In [10]: sns.histplot(df["sepal length (cm)"], kde=True)  
plt.show()
```



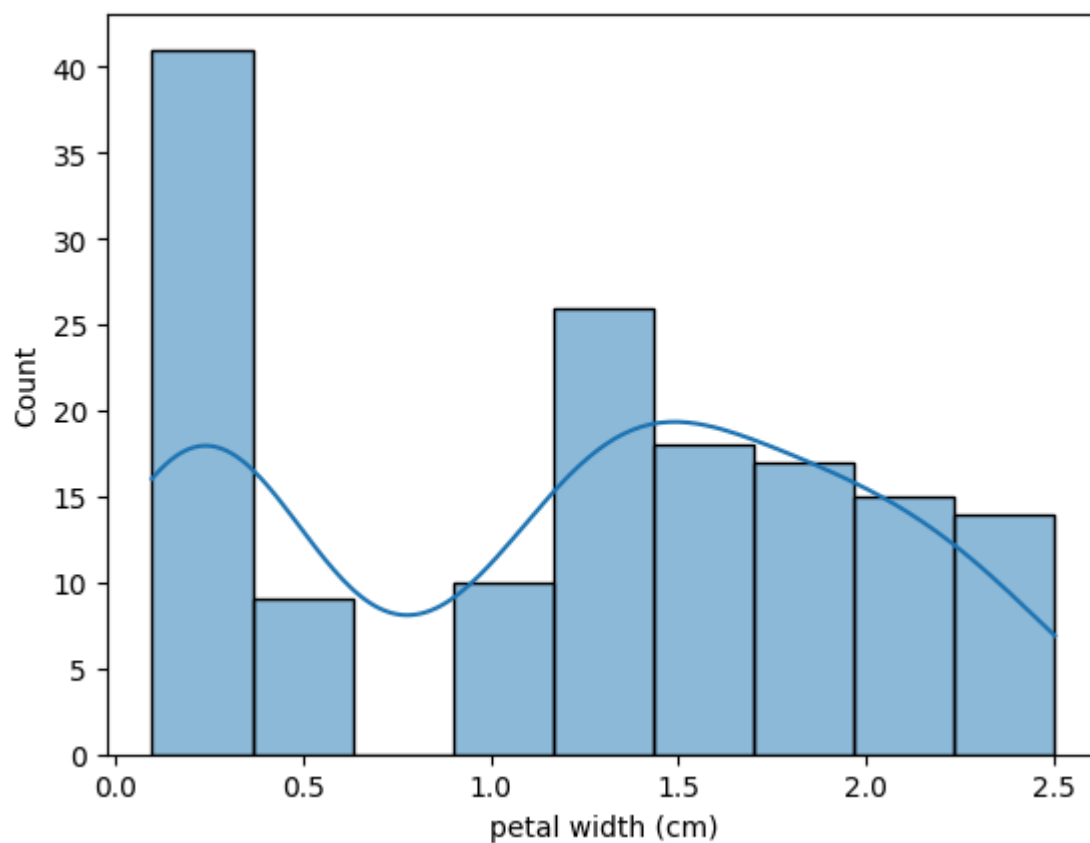
```
In [11]: sns.histplot(df["sepal width (cm)"], kde=True)
plt.show()
```



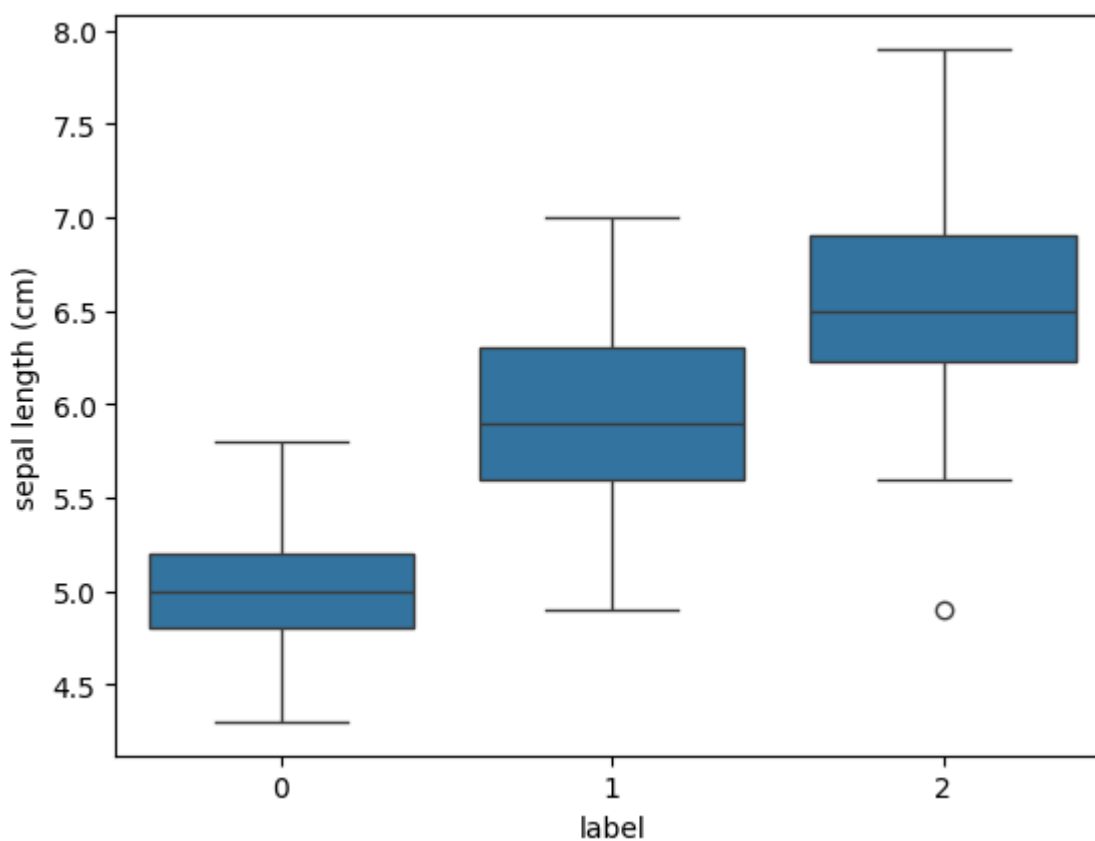
```
In [12]: sns.histplot(df["petal length (cm)"], kde=True)
plt.show()
```



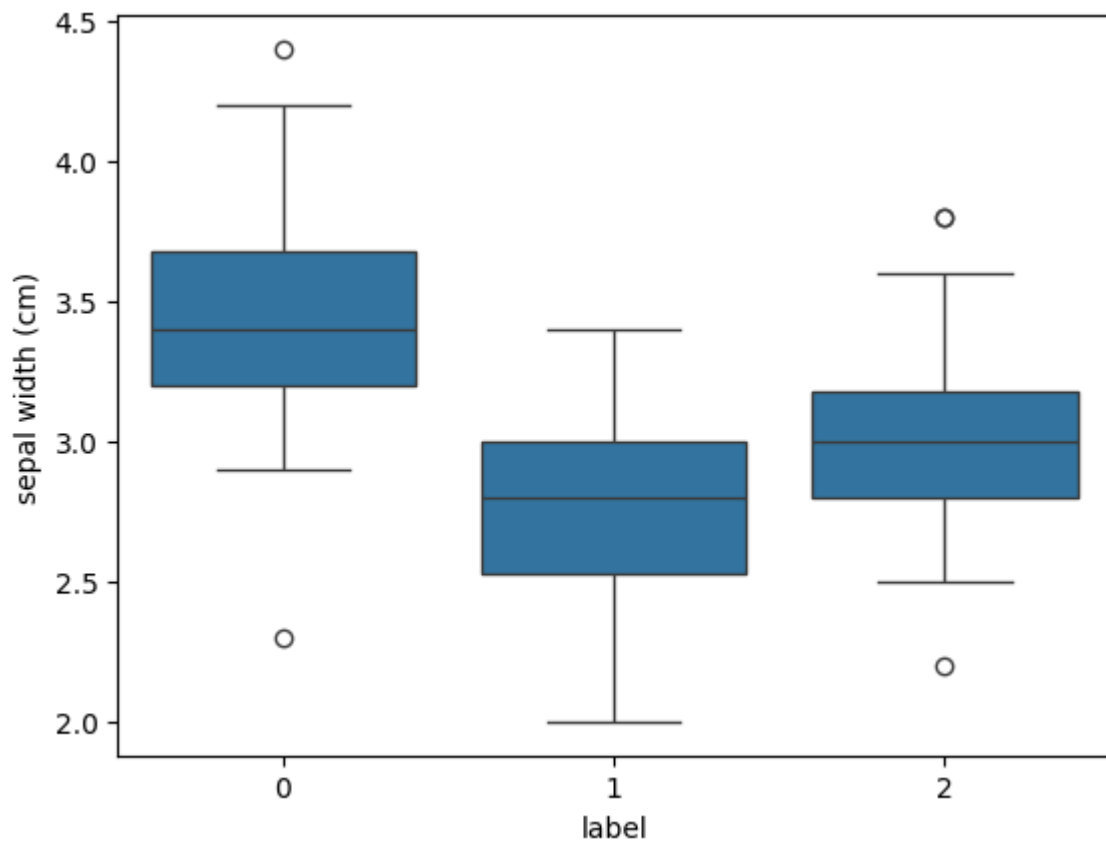
```
In [13]: sns.histplot(df["petal width (cm)"], kde=True)
plt.show()
```



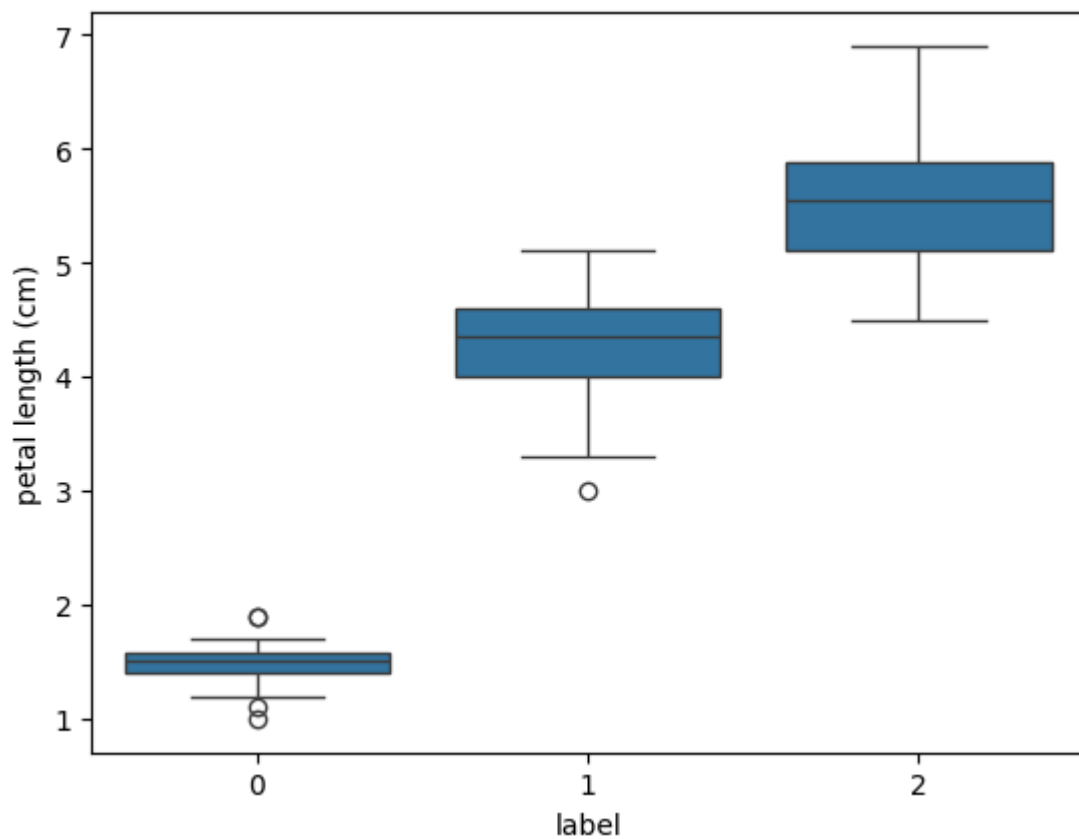
```
In [14]: sns.boxplot(x=df['label'], y=df["sepal length (cm)"])
plt.show()
```



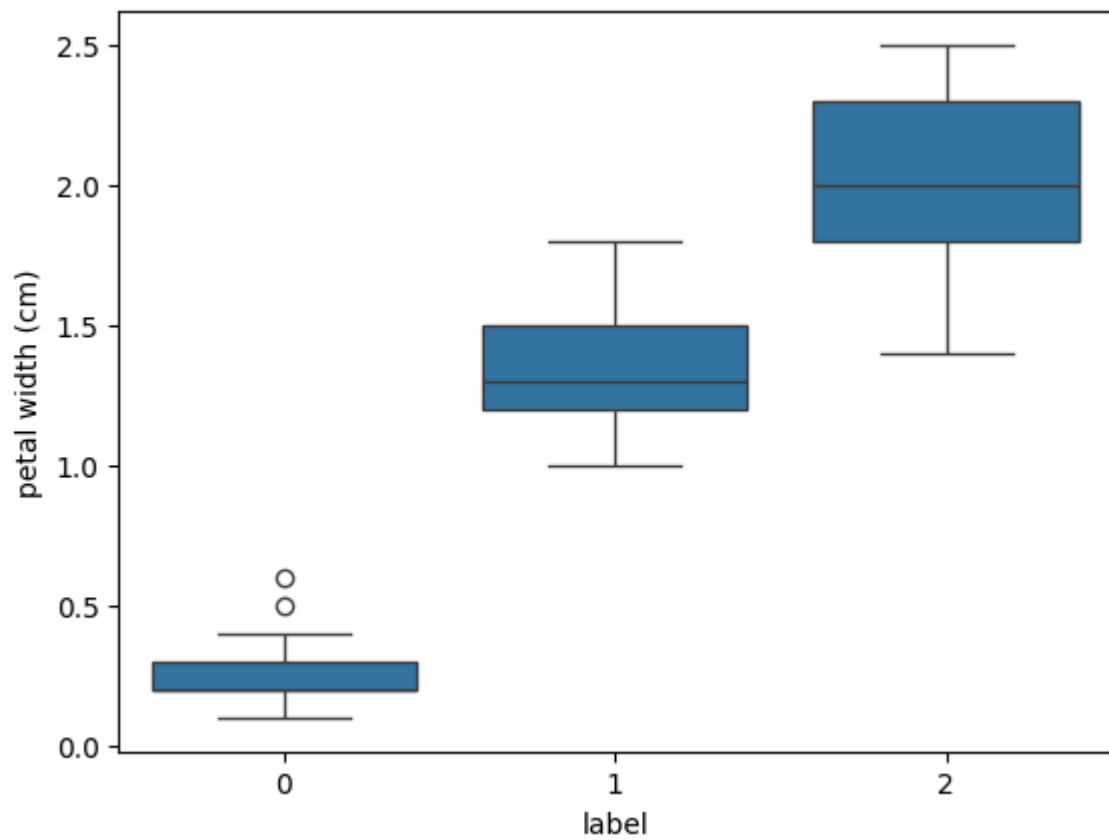
```
In [15]: sns.boxplot(x=df['label'], y=df["sepal width (cm)"])
plt.show()
```



```
In [16]: sns.boxplot(x=df["label"], y=df["petal length (cm)"])
plt.show()
```



```
In [17]: sns.boxplot(x=df['label'], y=df["petal width (cm)"])
plt.show()
```



```
In [ ]:
```